

Single-Board Microcontrollers

When you think of typical small microcontrollers, there are almost too many choose between:
https://en.wikipedia.org/wiki/Comparison_of_single-board_microcontrollers

Regarding three major chips in use today: the ATMEGA set, the RP2040, and the ESP32 set.

The **ATMEGA** chip is used on many Arduino models (Nano, Uno, Due, Mega modules) but it is now relatively slow and doesn't have a lot of coding space. You won't find many coded examples interfacing with camera modules, for example, since its visual processing would be relatively sluggish. There are thousands of popular projects online outlining how to interface with these devices with peripheral devices. Arduino has its own IDE, Integrated Development Environment.

The upgraded **RP2040** came out in 2021 and was a faster and cheaper option in comparison:
<https://www.utmel.com/components/rp2040-vs-atmega328-video-how-to-differentiate-them?id=1924>

This chip was popularized on the Raspberry Pi Pico, which costs about \$5 per module, but there are many other models available (<https://en.wikipedia.org/wiki/RP2040>). The RPi Pico W version has 26 GPIO pins and the ability to interface with Wifi or Bluetooth, in a fairly compact module.

The **ESP32** with Wifi & Bluetooth is basically the newer, faster version of the old ESP8266 Wifi-only chips (<https://makeradvisor.com/esp32-vs-esp8266/>). There are lots of models that use the ESP32 chip (<https://en.wikipedia.org/wiki/ESP32>) -- many have Wifi and Bluetooth, some don't. Typical prices are between \$5 and \$30 CAD per unit. The size ranges from tiny to large boards, from a couple of GPIOs to dozens, within this price range.

ESP32 chip devices will be the preferred MCU choice for TPJ.

Which ESP32 to Buy?

Selecting a model for your project depends on a few factors: budget, size constraints, number of GPIOs (in particular the number of Analog or Digital inputs or outputs, SPI, and I2C ports), ability to Wifi or Bluetooth, size of Flash RAM and PSRAM (memory or storage), and power.

Suppose you're searching **DigiKey.ca** amongst the RF Evaluation and Development Kits, Boards section: <https://www.digikey.ca/en/products/filter/rf-evaluation-and-development-kits-boards/>
You've found the latest ESP32-S3 models but there are many model options to choose from:

- DEVKITC-1-N8
- DEVKITC-1-N8R2
- DEVKITC-1-N8R8
- DEVKITC-1U-N8
- DEVKITC-1-N32R8V
- DEVKITM-1-N8 ...

What's the difference? The fastest way to differentiate between them is the check the datasheet: https://www.espressif.com/sites/default/files/documentation/esp32-s3-wroom-1_u_datasheet_en.pdf

On page 3, find the Flash and PSRAM differences between the suffix (N8, N8R2, N8R8, etc.). Also, there is the definition between 1 and 1U, where the "1" has a PCB antenna and the "1U" comes with an external antenna connector. Which one will you need?

To understand the differences between SPI standards, including XSPI, QSPI, Octal SPI, and PSRAM, first consider your project requirements and see if it ultimately matters for you: <https://www.arasan.com/blog/understanding-serial-peripheral-interface-spi-standards-a-comprehensive-overview/>

Sayal may be selling older models that use the ESP8266 chip instead of ESP32. These boards are often half the price but run slower and can't connect Bluetooth devices. You must read the tiny letters on the installed ESP chip itself to ensure which one is installed on these modules.

Amazon.ca offers many options! Make sure you understand which model you're really buying. For example, if you buy a NodeMCU model, does it have the ESP8266 chip installed (maybe from 2013), or the newer ESP32-C3-32S chip? Lots of different models have the "DevKit" name too.

Notes about PCB Design

The form factor of the board may matter to your project. Typical ESP32 boards are on DIP-Wide format, where the two rows of pins are set 0.6" apart. Is it DIP-40? DIP-32? DIP-28? DIP-16? These are also known as CDIP, which stands for Ceramic Dual In-line Package. In comparison, the typical ICs from DGS266 and other courses in your kit probably have 0.3" wide DIP settings.

Where to find it in PCB Design software? In Eagle, search for *dil* or look in the ic-package.lbr – IC Packages and Sockets. Alternatively, you may be able to download the EDA/CAD model package directly from the specific socket bought from DigiKey.ca:

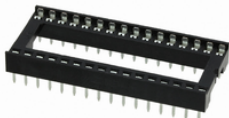


Image shown is a representation only. Exact specifications should be obtained from the product data sheet.



1-2199300-2

DigiKey Part Number	A120356-ND
Manufacturer	TE Connectivity AMP Connectors
Manufacturer Product Number	1-2199300-2
Description	CONN IC DIP SOCKET 32POS TINLEAD
Manufacturer Standard Lead Time	14 Weeks
Detailed Description	32 (2 x 16) Pos DIP, 0.6" (15.24mm) Row Spacing Socket Tin-Lead Through Hole
Customer Reference	
Datasheet	 Datasheet
EDA/CAD Models	 1-2199300-2 Models

Quick note about **battery life**: You can run ESP32 boards from a battery. We'll discuss this more in the lecture. Here are coding tips on how to extend battery life for ESP32 projects: <https://youtu.be/4EZYYPmQnJY> <https://youtu.be/IcLsuaEoHu0>

Videos on Connecting and Coding an ESP32 Board

Visual Studio Code is the preferred IDE of choice for TPJ.

You're going to see lots of videos online that are great tutorials for the Arduino IDE, which uses the Arduino framework. (That's when you use `#include Arduino.h` and the "setup()" and "loop()" framework in your code.) **Don't use Arduino IDE** for TPJ. Instead, use your time to develop skills with development software that you will more likely see in the workplace, like VS Code.

For example, **VS Code** can run still an Arduino platform, particularly if you install and use "PlatformIO" extension, but it can also run the ESP-IDF (IoT Development Framework) and you can go back to using our preferred "main()" framework. There are lots of resources online but they are a little trickier to track down, so we've compiled a few links to help you get started.

The following links are aimed at beginners learning to interface VS Code with ESP32 boards.

VS Code & ESP-IDF

ESP32 - Getting Started with ESP-IDF using VS Code [Easiest Method]

<https://youtu.be/5IuZ-E8Tmhg>

By: Yuri R (@YuriR)

The first video in a series about connecting your ESP32 board with VS Code and ESP-IDF.

ESP32 – How to Create Your First ESP IDF Project (From Scratch)

<https://youtu.be/oHHOCdmLiII>

By: Yuri R (@YuriR)

Another great video describing how to program in a non-Arduino framework, using ESP-IDF.

DevCon23 - ESP-IDF Getting Started (Beginners Guide to Key Concepts and Resources)

<https://youtu.be/J8zc8mMNKtc>

By: Espressif Systems (@EspressifSystems)

A summary of key concepts for beginners with limited programming/embedded experience.

VS Code & PlatformIO

PlatformIO: All You Need to Know in 10 Minutes

https://youtu.be/PYSy_PLjytQ

By: J's e-shack (@jse-shack825)

This will convince you to use VS Code and PlatformIO if you're still considering Arduino IDE.

Getting Started with ESP32 – Step-By-Step Tutorial

<https://youtu.be/tc3Qnf79Ny8>

By: Tomasz Tarnowski (@tomasztarnowski4434)

Steps to connect your MCU to the computer, write code in VS Code + PlatformIO, and deploy it.

Connect ESP32 to WiFi – Step-By-Step Tutorial

<https://youtu.be/aAG0bp0Q-y4>

By: Tomasz Tarnowski (@tomasztarnowski4434)

Tutorial showing how to connect your ESP32 to a Wifi network using VS Code + PlatformIO.

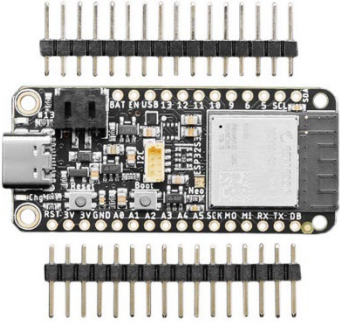
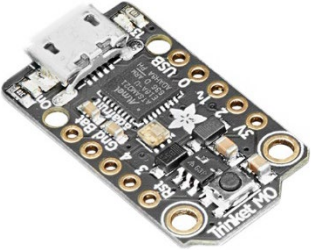
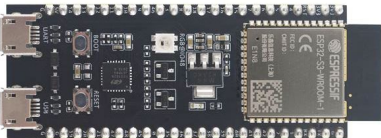
ESP-NOW - Peer to Peer ESP32 Network

<https://youtu.be/bEKjCDDUPaU>

By: DroneBot Workshop (@Dronebotworkshop)

Steps to create a private peer to peer network using the radio transceiver, no Wifi needed.

Example ESP32 Boards for sale on DigiKey.ca (2024)

	DigiKey P/N: 1528-5323-ND Manufacturer: Adafruit Industries LLC Description: ESP32-S3 FEATHER 8MB/0MB PSRAM Details: ESP32-S ESP32-S3 Transceiver; 802.11 b/g/n (Wi-Fi, WiFi, WLAN), Bluetooth® 5 2.4GHz Evaluation Board
	DigiKey P/N: 1597-113991114-ND Manufacturer: Seeed Technology Co., Ltd Description: SEEED STUDIO XIAO ESP32S3 WIFI/B Details: ESP32-S Transceiver; Bluetooth® 5 2.4GHz Evaluation Board
	DigiKey P/N: 2221-S007-PIN254-ND Manufacturer: M5Stack Technology Co., Ltd. Description: M5STAMPS3 WITH 2.54 HEADER PIN Details: ESP32-S3FN8 UIFlow ESP32-S Xtensa® MPU Embedded Evaluation Board
	DigiKey P/N: 1528-2361-ND Manufacturer: Adafruit Industries LLC Description: TRINKET M0 ATSAMD21E18 EVAL BRD Details: ATSAMD21E18 Trinket M0 SAM D ARM® Cortex®-M0+ MCU 32-Bit Embedded Evaluation Board
	DigiKey P/N: 1965-ESP32-S3-DEVKITC-1-N32R8V-ND Manufacturer: Espressif Systems Description: ESP32-S3-WROOM-2-N32R8V DEV BRD Details: ESP32-S ESP32-S3-WROOM-2-N32R8V Transceiver; 802.11 b/g/n (Wi-Fi, WiFi, WLAN), Bluetooth® 5 2.4GHz Evaluation Board